Raina Bethea Spring 26' Mid Project Report

Market Mood
Discovering Stock Market Regimes Using Unsupervised Data Mining

Introduction:

Financial markets do not behave consistently over time. Periods of steady growth and relatively low volatility can be followed by periods of declining prices, increased uncertainty, or rapid market movement. These different conditions are commonly described as market regimes. Identifying these regimes can help analysts better understand how market behavior changes over time and how different financial indicators behave under varying market conditions.

Traditional market analysis often relies on predefined thresholds or previously labeled market periods. However, data mining techniques provide an opportunity to identify patterns directly from historical financial data. In particular, unsupervised learning methods can group observations according to similarities in their characteristics without requiring each observation to be assigned a known market condition beforehand.

This project proposes an unsupervised data mining approach for discovering market regimes using historical U.S. stock market data. Daily market observations will be transformed into features representing characteristics such as returns, rolling returns, volatility, and momentum. Clustering algorithms will then be used to determine whether naturally occurring groups appear in the data and whether those groups represent meaningful market conditions. The resulting clusters will be analyzed and visualized across time to determine how effectively they characterize changes in market behavior.

Current Project Status:

The goal of *Market Mood* is to investigate whether unsupervised data mining techniques can identify distinct and interpretable market regimes from historical U.S. stock market behavior. Rather than assigning predefined labels such as growth, decline, or high-volatility periods, the project uses clustering to identify groups based on similarities in market characteristics.

The project currently uses historical daily S&P 500 closing values obtained from the Federal Reserve Bank of St. Louis (FRED). The raw dataset initially contained 1,305 observations covering October 2021 through October 2026. During preprocessing, 50 observations with missing S&P 500 values were removed, leaving 1,255 valid market observations.

Several behavioral features were then engineered from the closing-price data:

- Daily percentage return
- 5-day return
- 20-day return
- 20-day rolling volatility
- 20-day momentum

Momentum was calculated by comparing the current S&P 500 value with its 20-day moving average. Because rolling calculations require prior observations, rows without complete engineered features were removed. The resulting modeling dataset contains 1,235 complete observations.

Exploratory data analysis was then conducted to examine S&P 500 movement, rolling volatility, the distribution of daily returns, and relationships among the engineered variables. The analysis showed substantial variation in market behavior over time. Daily returns were generally concentrated near zero, while a smaller number of observations exhibited substantially larger positive or negative movements. Rolling volatility also showed several periods of elevated market variability.

A correlation analysis identified relationships among several return-based features. In particular, 20-day return and 20-day momentum had a correlation of 0.836, while 5-day return and 20-day momentum had a correlation of 0.750. Rolling volatility showed substantially weaker relationships with most return-based variables, suggesting that it provides a different dimension of market behavior.

Before clustering, the five selected behavioral features were standardized using “StandardScalar” so that differences in their numerical scales would not disproportionately influence the distance calculations used by K-Means.

Results:

Initial K-Means experiments evaluated values of *k* ranging from 2 through 6. Cluster quality was evaluated using both within-cluster inertia and silhouette score.

```
k = 2: Inertia = 4131.71, Silhouette Score = 0.369
k = 3: Inertia = 3440.12, Silhouette Score = 0.313
k = 4: Inertia = 2967.99, Silhouette Score = 0.244
k = 5: Inertia = 2612.56, Silhouette Score = 0.256
k = 6: Inertia = 2384.55, Silhouette Score = 0.257
```

Among the configurations tested, $k = 2$ produced the highest silhouette score at 0.369. Although inertia continued to decrease as additional clusters were introduced, the silhouette results suggest that two clusters currently provide the clearest separation among the tested configurations.

The two-cluster solution produced groups containing 900 and 335 observations. Their average characteristics were:

```
Cluster characteristics (%):
         daily_return  return_5d  return_20d  volatility_20d  momentum_20d
cluster
0                0.30        1.09        2.66            0.90          1.52
1               -0.64       -2.09       -3.79            1.19         -2.61
```

The clusters demonstrate clear differences in their market characteristics. Cluster 0 contains approximately 72.9% of the observations and is associated with positive returns, positive momentum, and lower average volatility. Cluster 1 contains approximately 27.1% of the observations and is associated with negative returns, negative momentum, and higher average volatility.

Based on these characteristics, the clusters have been given the preliminary interpretations Growth / Stable for Cluster 0 and Decline / Stress for Cluster 1. These labels were assigned only after examining the clustering results rather than being defined before clustering.

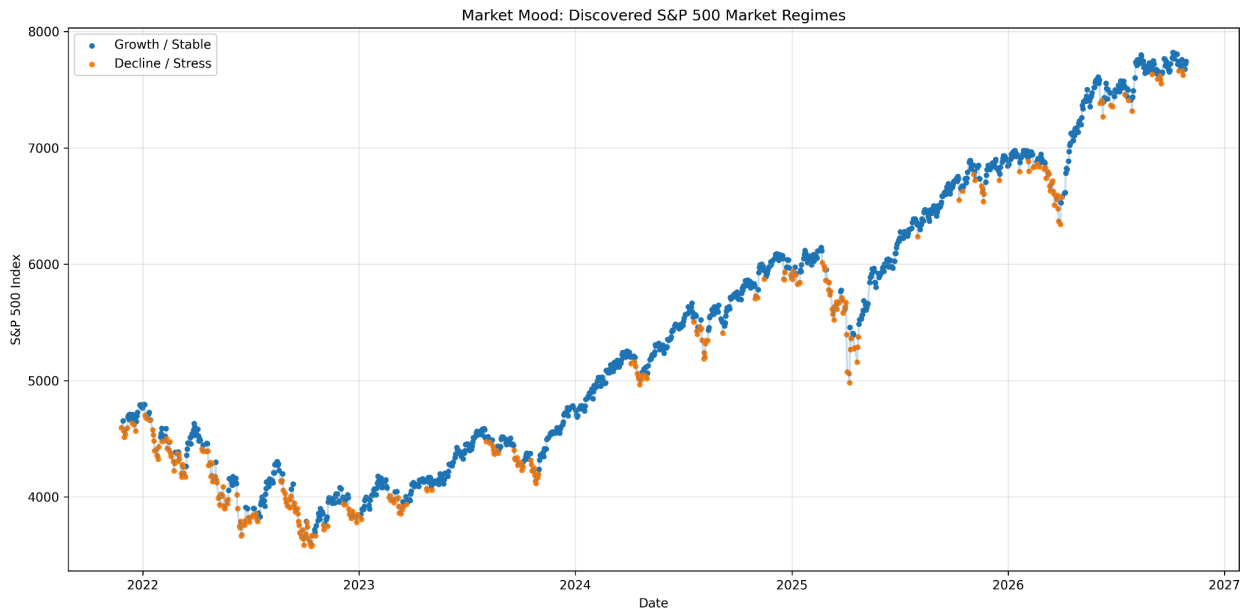


Figure 1. Market Mood: Discovered S&P 500 Market Regimes

Plotting the cluster assignments across the historical S&P 500 timeline provides additional support for their interpretation. Observations assigned to the preliminary Decline / Stress regime are concentrated around visible market declines and periods of instability, while Growth / Stable observations are more common during sustained upward market movement.

These results provide preliminary evidence that unsupervised clustering can identify distinct and interpretable patterns in the engineered market data. However, the two-cluster solution is not yet considered the final model and will be compared with additional experiments before final conclusions are made.

Progress vs. Milestones:

The project is generally following the development process established in the original proposal. The initial development stage included data collection, preprocessing, exploratory analysis, and feature engineering. All of which have now been completed.

Initial clustering experiments and analysis have also begun. K-Means has been evaluated across multiple values of k, preliminary cluster-quality metrics have been calculated, and the initial two-cluster solution has been interpreted and visualized across the S&P 500 timeline.

Therefore, the project has progressed from initial data preparation into the analysis and testing stage. The current implementation provides a working baseline that can now be expanded through additional experiments and comparison methods.

Future Work:

The next stage of the project will focus on evaluating the robustness and interpretability of the preliminary clustering results.

Because exploratory analysis identified relatively strong correlations among some return and momentum features, additional experiments will investigate whether removing highly correlated features changes cluster quality or produces more distinct regimes. The resulting silhouette scores, cluster characteristics, and regime assignments will be compared with the initial five-feature K-Means model.

Hierarchical clustering will also be investigated as an alternative unsupervised method. Its resulting clusters will be compared with the K-Means results to determine whether different clustering approaches identify similar structures within the market data.

Additional evaluation will focus not only on numerical clustering metrics but also on the financial interpretation of the discovered groups. The final analysis will examine when different regimes occur, how their return, volatility, and momentum characteristics differ, and whether the

identified regimes consistently correspond to visibly different patterns in historical market behavior.

The final stage will refine the visualizations, document the experimental comparisons, complete the source-code documentation and README instructions, and prepare the final report and project demonstration.